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REVIEW ARTICLE

Concussion recovery time, return-to-play protocols, and mandated sit-out periods among adult amateur contact field sports: A systematic review

Laura C. Power[1,2,3] · Ian C. Kenny[1,2,3,4] · Aaron J. Zynda[5] · John J.E. Mulvihill[2,3,6] · Michael W. Collins[5] · Anthony P. Kontos[5] · Tom M. Comyns[1,2,3]

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Abstract

Background Field-based contact sports, including American football and rugby are among those with the highest sport-related concussion (SRC) rates. Regarding SRC treatment, gradual return-to-play (RTP) protocols are commonly used, however variability in RTP protocols between sporting bodies and scholastic institutions makes accurately interpreting published SRC severity data difficult.

Aims Compare SRC recovery times, RTP protocols, and mandated sit-out period duration among adult amateur contact field sports.

Methods MEDLINE, CINAHL Complete, SPORTDiscus, APA PsycInfo, and APA PsycArticles were searched on May 13th, 2024. Eligibility criteria included (1) published in English in a peer-reviewed journal from 2015 to 2024, (2) adult population, (3) amateur, field-based contact sport athlete population, (4) diagnosed and/or suspected SRC, and (5) recovery time (days/weeks) data available.

Results 19/6085 (< 1%) studies comprising 58 datasets met all inclusion criteria. The studies comprised American football (10/19, 52.6%), field-hockey (5/19, 26.3%), lacrosse (9/19, 47.4%), ladies gaelic football (1/19, 5.3%), rugby (8/19, 42.1%), soccer (10/19, 52.6%), and sprint football (1/19, 5.2%). Mandated sit-out periods ranged from 0 to 21 days across sports, and RTP protocol durations ranged from 5 to 21 days. The most common RTP protocol duration was 7 days ($n = 31/58$), and mandated sit-out period was no same-day return ($n = 25/58$). Field hockey had the longest reported SRC recovery duration (34 days), and rugby had the longest mandated sit-out period (21 days).

Conclusion Recovery time following concussion in contact sports should be interpreted in light of sport type, mandated sit-out policies, and RTP protocols. Empirical evidence should drive decisions in RTP protocols and sit-out periods, which are currently inconsistent between sports.

Keywords Return-to-play · Sports injury · Sports medicine · Sports-related concussion

✉ Laura C. Power
power.laura@ul.ie

1 Department of Physical Education and Sport Sciences, University of Limerick, Limerick, Ireland

2 Health Research Institute, University of Limerick, Limerick, Ireland

3 Sport & Human Performance Research Centre, University of Limerick, Limerick, Ireland

4 Lero, The Research Ireland Centre for Software, Limerick, Ireland

5 Department of Orthopaedic Surgery, University of Pittsburgh, University of Pittsburgh Medical Center Sports Medicine Concussion Program, Pittsburgh, United States

6 School of Engineering and Bernal Institute, University of Limerick, Limerick, Ireland

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Introduction

Sport participation is associated with injury risk, in particular, sport-related concussions (SRC), with approximately 3.8 million SRCs reported annually in the United States [1]. An SRC is defined as "a traumatic brain injury caused by a direct blow to the head, neck, or body resulting in an impulsive force being transmitted to the brain" which initiates a neurometabolic cascade and results in a range of clinical signs (e.g., loss of consciousness, amnesia), symptoms (e.g., headache, dizziness), and impairments (e.g., neurocognitive, vestibular/ocular motor) [2, 3]. Recovery from SRC usually takes up to one month; however, 10–15% of patients may experience persisting symptoms [3–5]. Risk factors for persisting symptoms and prolonged recovery include pre-injury factors, such as female sex and concussion history, and post-injury factors, such as higher initial symptom severity and continued play [6–8]. In addition to these established risk factors, sport-specific factors, such as sport type, contact level, level of competition, and return-to-play (RTP) protocols, may also alter recovery time. Understanding how sport-specific factors influence recovery following SRC may inform targeted management and treatment plans, improve prognostication, shift attitudes and perceptions around SRC recovery, and help bridge the gap between research and policy.

Previous research has consistently demonstrated greater concussion incidence and risk in contact sports (e.g., football, rugby) compared to limited (e.g., baseball, volleyball) and non-contact (e.g., cross country, swimming) sports [9, 10], warranting a substantial research focus. Field-based contact sports, including, but not limited to, American football, rugby, Gaelic football, and lacrosse, are among those with the highest concussion rates [11–13]. Level of competition also plays a role, although research is mixed, with evidence demonstrating greater concussion rates in amateur athletes than professional athletes [14]. Despite extant evidence examining concussion risk based on these sport-specific factors, less is known about differences in recovery time. Liebel et al. found a median recovery range of 11–16 days across various sports [15]. American football was in the highest third of recovery times, and lacrosse was in the lowest third, despite a similar contact level and injury risk categorisation [15]. Meanwhile, other research found no statistical significance between sport type and SRC recovery duration [16]. In a systematic review of the time to return to sports, the mean recovery range was 6.7–34.7 days for amateur athletes, but the authors did not examine the recovery of amateur athletes within contact sports [9]. Altogether, the salience of concussion in contact field sports at the amateur level and mixed evidence on recovery times following SRC warrants further review.

The variability of return-to-play (RTP) protocols within different sports and organisations adds to the complexity of assessing recovery time. The Concussion in Sport Group (CISG) consensus statement supports a gradual RTP approach following SRC with six progressive stages, including 24-hour breaks between each stage [3]. Therefore, athletes can expect at least one week out from sports participation before unrestricted return [17], although SRC is heterogeneous by nature and progression through the protocol varies [18]. Since the introduction of RTP protocols, time to unrestricted return-to-sport has quintupled [3]. The CISG consensus statement, however, highlights that it is to be used as a guide and is not a ''clinical practice directive or legal standard of care'' [3]. Many sports organisations have released their own concussion management guidelines, with variations of initial rest phases, RTP steps, sport-specific recommendations, and some enforcing a mandated sit-out period, such as the Irish Rugby Football Union with a 21-day mandated adult sit-out [19, 20]. International policies' minimum time to RTP ranges from 5 to 30 days [20]. Therefore, it is difficult to clearly understand differences between sport recovery durations, particularly between studies, without awareness of specific mandated sit-out time enforcement or variations between sporting codes RTP protocol duration. These sit-out periods and variations in RTP protocols may skew concussion recovery data, potentially misinforming clinicians of expected RTP times for their athletes and negatively impacting clinical decision-making.

This systematic review of the extant literature aims to: (1) compare SRC recovery times in adult amateur athletes across contact field sports, (2) compare intra-sport concussion recovery time variability across studies, and (3) describe and analyse each sport's recommended concussion medical clearance policy (i.e., RTP protocol, mandatory sit-out period). We hypothesise that sports with longer durations of medical clearance policies will result in longer SRC recovery duration.

Methods

This systematic review of the literature was done in accordance with the PRISMA checklist [21] and PERSiST guidelines [22].

Information sources

Database sources were last searched on May 13th, 2024, through EBSCOhost Web (EBSCO Information Services, 2024). Database selection was limited to MEDLINE, CINAHL Complete, SPORTDiscus, APA PsycInfo, and

APA PsycArticles. Reference lists were screened to supplement the electronic database findings.

Search strategy

A librarian from the institution was consulted to help formulate a search strategy and enhance report quality [23]. We searched for S1 ('sports' MEsH OR 'return to sport' MeSH), S2 in Title or Abstract ('sport*' OR 'return to sport*' OR 'return to play' OR 'return to activit*'), S3 ('Brain Concussion' MeSH OR 'Brain Injuries, Traumatic' MeSH), S4 in Title or Abstract ('brain concussion' OR 'traumatic brain injur*' OR 'concussion'), S5 (S1 OR S2), S6 (S3 OR S4) and S7 (S5 AND S6). The full search strategy can be found in Supplemental Material 1. Magazines and continuing education units were filtered out through EBSCOhost.

Eligibility criteria

Studies were included if they: (1) were published in a peer-reviewed journal within the last 10 years (2015–2024), (2) were published in the English language, (3) comprised an adult population (≥18 years old), (4) sampled amateur, field-based contact sport athletes, and (5) reported concussion recovery data (in the format of days/weeks). In the case of multiple publications from the same dataset, only the relevant paper was included in this study. Studies were also excluded if there were (1) concussion recovery time data in the format of number of trainings/games missed only, (2) no concussion recovery time metric at all, (3) less than 75% of the data were SRCs, or (4) systematic reviews or meta-analyses.

Selection process

All papers retrieved from the search were exported to Clarivate EndNote 21 [24] and Covidence Systematic Review Software [25]. Title and abstract screening were completed on Covidence by a single reviewer. Two reviewers independently conducted full-text reviews of articles following eligibility criteria, with a third reviewer available for any conflict resolution.

Data collection process

Data were extracted from selected papers to be stored in Microsoft Excel by a single reviewer. Where data for concussion recovery time per sport (e.g. multi-sport studies) were unavailable, corresponding authors were emailed with a supplemental data request. Twenty-one authors were contacted, with 7 authors supplying supplemental data to be included in the review. For RTP policy acquirement, should the policy not be available within published literature or the sporting organisation's online systems, the organisations were contacted through email by the first author for a copy of the active policy during the paper's data-collection period (Fig. 1).

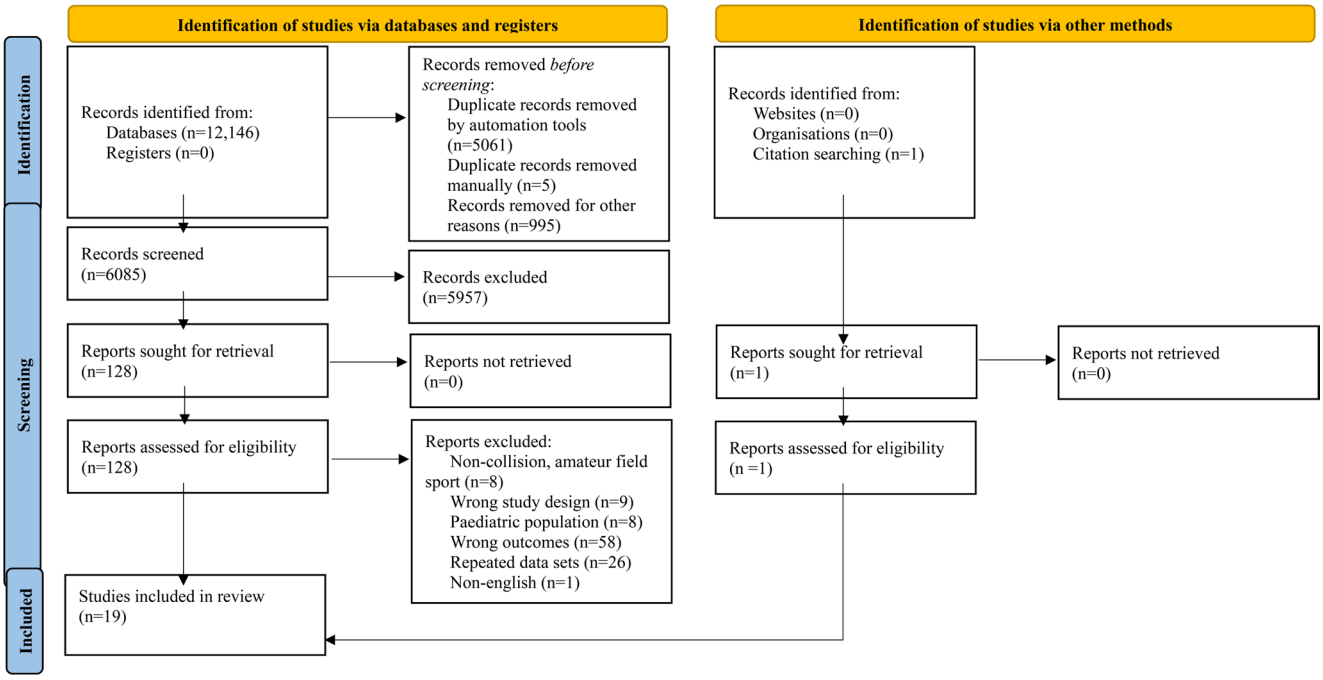

Fig. 1 PRISMA flow diagram [21]

Data items and definitions

Data collected included concussion recovery time (day/months), sex, contact field sport, country, corresponding concussion RTP policy at the time of data collection (where available), setting (e.g., match, training), number of concussions, and concussion incidence.

Concussion recovery time was defined in this study as the time between injury incidence and return to unrestricted play (training or match).

Sport contact levels were selected using previously published contact sport categorisation [26] used in published concussion research [27].

Collegiate athletes participate in varsity or club sports within a university, college, or other third-level institution.

Match and/or *game* were used interchangeably, alongside *training* and/or *practice*.

Return-to-play protocols were described as the amount of time spent in the gradual RTP procedure, either outlined in the journal article, citations, or the corresponding organisation's online material.

Mandated sit-out periods were described as an amount of time whereby the athlete/player must not under any circumstance RTP; these were identified and interpreted by the first author by use of strict language such as 'must not,' 'will not,' and 'cannot,' rather than suggestive language such as 'should' or 'are advised to.'

Risk of bias assessment and certainty assessment

The National Institute of Health Quality Assessment Tool for Observational Cohort, Cross-Sectional Studies, and Case-Control Studies was used by a single reviewer to assess the risk of bias [28]. The Strength of Recommendation Taxonomy (SORT) was used by a single reviewer to assess certainty in the body of evidence [29] (Table 1).

Table 1 Summary of the National Institute of health quality assessment tool for observational Cohort, Cross-Sectional Studies, and Case-Control Studies, alongside the strength of recommendation taxonomy (SORT)

Paper	Quality Assessment[1]	Level of Evidence[2]
Asken et al.[32]	Good	2
Ball et al.[41]	Good	3
Bayram et al.[33]	Fair	3
Beidler et al.[45]	Fair	2
Bretzin et al.[35]	Fair	2
Buckley et al.[44]	Good	2
Covassin et al.[36]	Good	2
Davis-Hayes et al.[16]	Good	2
Fuller et al.[42]	Good	2
Gallagher et al.[40]	Good	2
Howell et al.[30]	Good	2
Leahy et al.[37]	Fair	3
Liebel et al.[15]	Good	2
Roberts et al.[43]	Fair	2
Sterling et al.[31]	Fair	2
Wang et al.[8]	Fair	3
Wasserman et al.[34]	Fair	3
Yener at al.[38]	Good	3
Yeomans et al.[39]	Good	2

1. The National Institute of Health Quality Assessment Tool for Observational Cohort, Cross-Sectional Studies/Case-Control Studies

2. The Strength of Recommendation Taxonomy

Results

Descriptive findings

Sport-specific concussion and recovery time results

American football A total of 11/19 (58.0%) studies involved American football. The number of SRCs reported ranged from 23 to 708 across data collection periods ranging from one to 16 seasons [30, 31]. The RTP durations ranged from 7 days [32] to 16 days [30]. Only one of the American football studies reported females within the sample, totalling 0.7% female representation [33]. Davis-Hayes et al. (2017) reported no difference in RTP times for American Football versus any other organised sports investigated, where the mean RTP for all sports combined was found to be 14 (± 17) [16]. Sterling et al. (2023) reported that 589/708 (83.2%) concussions were categorised as simple (0–10 days time-loss), whereas 119/708 (16.8%) were complex (> 10 days time-loss). In a study of American football players in the UK, Bayram et al. (2023) reported that approximately one in three student-athletes sustained at least one concussion during the season [33]. These SRCs accounted for 5.8% of all severe injuries (time-loss of more than four weeks) [33]. Wasserman et al. (2016) reported that only 6.1% of American football players with a concussion took 28 days or longer to recover, with the majority taking '1–6 days' and '7–13 days' [34].

Field-Hockey A total of 5/19 (26.3%) studies involved field-hockey. The number of SRCs reported ranged from 3 to 59 [8, 35]. Concussion RTP median times ranged from 10 to 37 days [8, 30]. It should be noted that the study with the highest RTP median had a small sample size with $n = 3$; therefore, outliers may have skewed the median RTP duration [8]. Wasserman et al. (2016) found that most SRC RTP

occurred within 7–13 days for this population, with 20% taking over 28 days to recover [34].

Lacrosse A total of 9/15 (47.4%) studies involved lacrosse, resulting in 13 data sets where male and female data were analysed separately. Median SRC RTP time for the mixed-sex population was 24 days [30], with male-only data remaining consistent across two studies with results of 14 and 13.9 days (IQR 12.1–17.1) [15, 35]. Female-only median data ranged from 8 days (IQR 6–56) to 19 days (IQR 12.25–45.00.25.00) [8, 32]. One study reported mean RTP data for males (Match: 8.93 ± 7.34 days, Training: 8.70 ± 7.99 days) and females (Match: 6.16 ± 5.89 days, Training: 6.65 ± 3.89 days) [36]. Wasserman et al. (2016) found both male and female lacrosse players recovered most frequently within '7–13 days' [34].

Ladies Gaelic football A total of 1/19 (5.3%) studies involved ladies gaelic football.The median RTP time reported in this study was 7 days (IQR 17.3) [37].

Rugby A total of 8/19 (42.1%) studies involved rugby, resulting in 10 datasets in which male and female data were analysed separately. Within published median RTP data, one set of mixed-sex data was available (21.5 days (IQR 8–59.5.5)) [38]. For males, the median RTP was 21 [35, 39] (IQR 20–32) [39]. For females, the median RTP ranged from 18.5 days (IQR 16–19) to 34 days [35, 40]. Mean RTP data were reported for three male-only populations with a range of 11.8 to 25.2 days [41–43]. It should be noted that 5/8 (62.5%) studies researched rugby union data [38, 39, 41–43], whilst 3/8 (37.5%) studies did not specify which form of rugby data was included [30, 35, 40].

Soccer A total of 10/19 (52.6%) studies involved soccer, resulting in 17 datasets in which male and female data were analysed separately. Within published median RTP data, mixed-sex was 18 days [30]. For male-only data, median RTP ranged from 6.5 to 17 days [8, 35], while female-only median RTP findings ranged from 8 (IQR 6–9) to 15 days (IQR 13–17) [32, 44]. Covassin et al. (2016) found that, on average, males returned faster (Match: 8.27 ± 8.07 days, Training: 6.14 ± 6.15 days) than their female counterparts (Match: 7.95 ± 6.21 days, Training: 9.33 ± 8.54 days)) [36]. Wasserman et al. (2016) depict the largest RTP category duration as '1–6 days' for both male and female soccer players, with the '7–13 days' range being the second largest for both sexes [34].

Sprint football A total of 1/19 (5.3%) studies involved sprint football. The median RTP time was 14 days [35] (Table 2).

Relationship of RTP protocols and mandatory sit-out periods to recovery time

Fourteen different sources for RTP protocol variations were identified throughout this systematic review. All protocols consisted of a stepwise progression in RTP, aligning with the consensus guidelines [3]. These RTP protocol durations are available to view in Table 3, alongside the source of the protocol. Altogether 12/19 (63.2%) studies offered either direct or cited reference to their RTP protocol [8, 16, 30, 32, 35, 36, 38, 40, 41, 43–45], 4/19 (21.1%) made no reference to SRC RTP protocols [31, 34, 39, 42], 2/19 (10.5%) investigated RTP compliance within the study [15, 37], and 1/19 (5.3%) stated that there was no RTP management policy available at time of study [33]. RTP protocol duration ranged from five to 21 days [39, 40]. Some policies [3, 4, 46–48] may have differed between institutions [35] as they act as guidelines for a policy's framework and clinical practice [46]. Rugby union had the longest RTP protocol of all included sports, at 21 days [39] (Table 1).

Studies identified that mandated sit-out periods varied, ranging from no same-day return to 21 days [39, 48]. Rugby and ladies gaelic football were the only two sports with clear displays of extended mandated sit-out periods of over one day/no same-day return, ranging from 11 to 21 days [30, 37–39, 41–43]. Rugby union was found to have the longest mandated sit-out period of 21 days [49]. Rugby studies had mandated sit-out periods of more than one day in 6/8 (75.0%) of the studies (Table 2). The authors found that the duration of 'mandated' sit-out time was one day or 'no same-day return' for 91% (n = 10/11) of U.S. collegiate athlete studies, while Gallagher et al. (2018) was an outlier identifying 6 days minimum RTP, which the authors deemed mandatory [40] (Table 3).

In relation to compliance with RTP protocols and medical practitioner involvement, 10/19 (52.6%) studies reported medical involvement in the diagnosis, treatment, and/or clearance of SRC by a medical practitioner [8, 15, 16, 30, 32, 35, 36, 40, 44, 45], while 5/19 (26.3%) reported a medical practitioner conducted injury recording/logging [34, 38, 41–43]. A total of 2/19 (10.5%) studies reported participants' completion of an RTP protocol was part of the study inclusion criteria [44, 45], with 1/19 studies highlighting that the strict management protocols within the study may diminish the generalisability of findings [35].

Discussion

This systematic review supports variability in SRC recovery time across amateur adult contact field sports, with a median RTP between 6.5 and 37 days in male soccer and

Table 2 Summary of included papers

Author	Title	Year	Sport*	Population	Age (mean, in years)	Sample size** (*n*=)	Country
Asken et al.[32]	Playing Through It": Delayed Reporting and Removal From Athletic Activity After Concussion Predicts Prolonged Recovery	2016	Multi-sport	Collegiate Athletes (National Collegiate Athletic Association)	20.4 ± 1.3	97 participants	United States of America
Ball et al.[41]	Movement Demands and Injury Characteristics in Under-20-Years University Rugby Union Players	2020	Rugby Union	Collegiate Athletes (under 20 years old)	19.3 ± 0.6	34 participants	Australia
Bayram et al.[33]	Epidemiology of American Football Injuries at Universities in the United Kingdom	2020	American Football	Collegiate Athletes	21.3	410 participants	United Kingdom
Beidler et al.[45]	Racial identity and concussion diagnosis and recovery trajectories in collegiate athletes: a LIMBIC MATARS investigation	2023	Multi-sport	Collegiate Athletes (National Collegiate Athletic Association)	*Racial Identity Groups* *Black* 20.02 ± 1.4 *White* 19.87 ± 1.4	410 concussions	United States of America
Bretzin et al.[35]	Clinical Recovery Timelines After Sport-Related Concussion in Men's and Women's Collegiate Sports	2022	Multi-sport	Collegiate Athletes (Ivy-League)	*Male* 19.96 ± 1.3 *Female* 19.57 ± 1.2	1974 participants	United States of America
Buckley et al.[44]	Acute Physical and Mental Activity Influence on Concussion Recovery.	2022	Multi-sport	Collegiate Athletes (National Collegiate Athletic Association)	19.6 ± 1.4	78 participants	United States of America
Covassin et al.[36]	Sex Differences in Reported Concussion Injury Rates and Time Loss From Participation: An Update of the National Collegiate Athletic Association Injury Surveillance Program From 2004–2005 Through 2008–2009	2016	Multi-sport	Collegiate Athletes (National Collegiate Athletic Association)	-	1702 concussions	United States of America
Davis-Hayes et al.[16]	Sex-specific Outcomes and Predictors of Concussion Recovery	2017	Multi-sport	Collegiate Athletes (National College Athletic Association)	19 ± 1	1200 participants	United States of America
Fuller et al.[42]	Eight-season epidemiological study of injuries in men's international Under-20 rugby tournaments	2018	Rugby Union	International amateur (under 20 years old)	19.1 ± 0.82	3918 participants	United Kingdom
Gallagher et al.[40]	The Effects of Sex Differences and Hormonal Contraception on Outcomes after Collegiate Sports-Related Concussion.	2018	Multi-sport	Collegiate Athletes	*Male* 19.8 ± 1.2 *Female* 19.6 ± 1.1	90 participants	United States of America
Howell et al.[30]	Exercise in the First Week Following Concussion Among Collegiate Athletes: Preliminary Findings	2020	Multi-sport	Collegiate Athletes (National Collegiate Athletic Association)	*Group 1*: 20.1 ± 1.3 *Group 2*: 20.3 ± 1.3	72 participants	United States of America
Leahy et al.[37]	Concussion in Ladies Gaelic Football: Self-reported History, Clinical Profiles, and Management Behavior	2023	Gaelic Football	Amateur (Club) Adult Athletes	24.4 ± 5.6	523 participants	Ireland
Liebel et al.[15]	Sport-Specific Recovery Trajectories for NCAA Collegiate Athletes Following Concussion	2023	Multi-sport	Collegiate Athletes (National Collegiate Athletic Association (NCAA) and Department of Defense (DoD) CARE Consortium)	19 ± 1	1049 participants	United States of America

Table 2 (continued)

Author	Title	Year	Sport*	Population	Age (mean, in years)	Sample size** (*n*=)	Country
Roberts et al.[43]	Concussions and Head Injuries in English Community Rugby Union Match Play	2016	Rugby Union	Amateur (Club) Adult Athletes	25.6	2916 participants	England
Sterling et al.[31]	Effect of Mouthguard Type on Severity of Concussion During Game-Related College Football Competition. A 16-Season Prospective Study	2023	American Football	Collegiate Athletes (Football Bowl Subdivision conferences)	-	-	United States of America
Wang et al.[8]	Factors Associated With a Prolonged Time to Return to Play After a Concussion	2022	Multi-sport	Collegiate Athletes (National Collegiate Athletic Association)	-	159 participants	United States of America
Wasserman et al.[34]	Epidemiology of Sports-Related Concussions in National Collegiate Athletic Association Athletes From 2009–2010 to 2013–2014: Symptom Prevalence, Symptom Resolution Time, and Return-to-Play Time	2016	Multi-sport	Collegiate Athletes (National College Athletic Association Injury Surveillance Program)	-	1670 concussions	United States of America
Yener at al.[38]	The association between sport-related concussion and musculoskeletal injury in university rugby athletes	2022	Rugby Union	Collegiate Athletes	-	181 participants	England
Yeomans et al.[39]	Injury Trends in Irish Amateur Rugby: An Epidemiological Comparison of Men and Women	2021	Rugby Union	Amateur (Club) Adult Athletes	*Male* 25.1 ± 4.1 *Female* 27.2 ± 5.3	1193 participants	Ireland

*Multi-sport studies include analysis of various sporting types, typically in third-level settings in the United States of America such as American Football, Lacrosse, Soccer, Field-Hockey, Volleyball, Diving etc. Specifics on each sport may be seen in Tables 2 and 3

** Where reported, sample size was shown as number of participants. In the absence of this data, overall concussions were reported

Table 3 Summary of concussion Return-to-Play times per sport, the corresponding organisation's Return-to-Play protocol duration and the policy's mandated sit-out period (where relevant)

Sport	Author	Sex (Male/Female/Both)	Concussion return-to-play time	Organisation's return-to-play protocol duration (at time of study)*	Policy's mandated sit-out period (at time of study)
American Football	Asken et al. [32]	Male	7 days (median, IQR 6–9)	7 days• [4, 32, 47]	0–1 day (guidelines changed during study) • [4, 32, 47]
	Beidler et al. [45]	Male	11 days (median, IQR 8–19.5.5)[a]	7 days• [3, 45]	No same day return • [3, 45]
	Bretzin et al. [35]	Male	13 days (median)	7 days• [35, 46]	No same day return• [17, 35]
	Buckley et al. [44]	Male	12 days (median, IQR 9–16.5.5)	7 days• [4, 44, 47]	1 day • [4, 44, 47]
	Howell et al. [30]	Male	16 days (median)	7 days •[30, 46	No same day return• [30, 46]
	Liebel et al. [15]	Male	14 days (median, IQR 13.1–14.9)	7 days▫ [48]	No same day return (if diagnosed) ▫ [48]
	Wang et al. [8]	Male	14 days (median, IQR 11.75–18.25)	7 days ▫[48]	No same day return (if diagnosed) ▫ [48]
Field-Hockey	Bretzin et al. [35]	Female	13 days (median)	7 days • [35, 46]	No same day return• [17, 35]
	Howell et al. [30]	Female	10 days (median)	7 days •[30, 46]	No same day return •[30, 46]
	Wang et al. [8]	Female	37 days (median, IQR 34–43)	7 days▫ [48]	No same day return (if diagnosed) ▫ [48]
Lacrosse	Asken et al. [32]	Female	8 days (median, IQR 6–56)	7 days •[4, 32, 47]	0–1 day (guidelines changed during study) •[4, 32, 47]
	Bretzin et al. [35]	Male	14 days (median)	7 days• [35, 46]	No same day return• [17, 35]
	Bretzin et al. [35]	Female	16 days (median)	7 days• [35, 46]	No same day return• [17, 35]
	Covassin et al. [36]	Male	Match: 8.93 ± 7.34 days (mean) Training: 8.70 ± 7.99 days (mean)	-	-
	Covassin et al. [36]	Female	Match: 6.16 ± 5.89 days (mean) Training: 6.65 ± 3.89 days (mean)	-	-
	Gallagher et al. [40]	Female	14 days (median)	6 days [40]	6 days [40]
	Howell et al. [30]	Both	24 days (median)	7 days •[30, 46]	No same day return• [30, 46]
	Liebel et al. [15]	Male	13.9 days (median, IQR 12.1–17.1)	7 days▫ [48]	No same day return (if diagnosed) ▫ [48]
	Liebel et al. [15]	Female	11.2 days (median, IQR 9.9–13.1)	7 days▫ [48]	No same day return (if diagnosed) ▫ [48]
	Wang et al. [8]	Female	19 days (median, IQR 12.25–45.00.25.00)	7 days▫ [48]	No same day return (if diagnosed) ▫ [48]
Ladies Gaelic Football	Leahy et al. [37]	Female	7 days (median, IQR 17.3)	15 days▫ [61]	15 days▫ [61]
Rugby	Ball et al. [41]	Male	11.8 days (mean)	11 days [41, 62]	11 days [41, 62]
	Bretzin et al. [35]	Male	21 days (median)	7 days • [35, 46]	No same day return [17, 35]
	Bretzin et al. [35]	Female	34 days (median)	7 days• [35, 46]	No same day return [17, 35]
	Fuller et al. [42]	Male	21 days (mean) [b]	14 days [20, 63]	14 days [20, 63]
	Gallagher et al. [40]	Female	18.5 days (median, IQR 16–19)	6 days [40]	6 days [40]
	Howell et al. [30]	Female	26 days (median)	7 days •[30, 46]	No same day return •[30, 46]
	Roberts et al. [43]	Male	25.2 days missed [c] (median, 95% CI 22.5–28)	11 days [d][43]	19 days[d] [43]
	Yener et al. [38]	Both	21.5 days (median, IQR 8.0–59.5.0.5)	5 days[38, 64]	19 days[38]
	Yeomans et al. [39]	Male	21 days (median, IQR 21–28)	21 days•[49]	21 days• [49]
	Yeomans et al. [39]	Female	21 days (median, IQR 20–32)	21 days•[49]	21 days• [49]

Table 3 (continued)

Sport	Author	Sex (Male/Female/Both)	Concussion return-to-play time	Organisation's return-to-play protocol duration (at time of study)*	Policy's mandated sit-out period (at time of study)
Soccer	Asken et al. [32]	Female	8 days (median, IQR 6–9)	7 days [4, 32, 47]	0–1 day (guidelines changed during study) [4, 32, 47]
	Bretzin et al. [35]	Male	17 days (median)	7 days• [35, 46]	No same day return• [17, 35]
	Bretzin et al. [35]	Female	14 days (median)	7 days• [35, 46]	No same day return• [17, 35]
	Buckley et al. [44]	Male	9 days (median, IQR 8.5–9.5)	7 days• [4, 44, 47]	1 day• [4, 44, 47]
	Buckley et al. [44]	Female	15 days (median, IQR 13–17)	7 days• [4, 44, 47]	1 day• [4, 44, 47]
	Covassin et al. [36]	Male	Match: 8.27 ± 8.07 days (mean) Training: 6.14 ± 6.15 days (mean)	-	-
	Covassin et al. [36]	Female	Match: 7.95 ± 6.21 days (mean) Training: 9.33 ± 8.54 days (mean)	-	-
	Gallagher et al. [40]	Female	13 days (median)	6 days [40]	6 days [40]
	Howell et al. [30]	Both	18 days (median)	7 days •[30, 46]	No same day return •[30, 46]
	Liebel et al. [15]	Male	14.8 days (median, IQR 13.3–18)	7 days▫ [48]	No same day return (if diagnosed) ▫ [48]
	Liebel et al. [15]	Female	13.8 days (median, IQR 12.8–14.9)	7 days ▫ [48]	No same day return (if diagnosed) ▫ [48]
	Wang et al. [8]	Male	6.5 days (median, IQR 6.25–6.75)	7 days▫ [48]	No same day return (if diagnosed) ▫ [48]
	Wang et al. [8]	Female	10.5 days (median, IQR 7–14.25.25)	7 days▫ [48]	No same day return (if diagnosed) ▫ [48]
Sprint Football	Bretzin et al. [35]	Male	14 days (median)	7 days •[35, 46]	No same day return •[17, 35]

*Where return-to-play protocol specifics were not included within the article, referenced frameworks used or the organisation's corresponding protocol at the time of study were used in its place. All outsourced protocols and material are referenced here

• Depicts protocol duration and mandated return-to-play times source were cited in the manuscript and outsourced from citations, relied on interpretation from authors (may have differed slightly between different institutions within multi-site studies or clinicians' interpretation/execution of the protocol)

▫ Depicts protocol duration and mandated return-to-play times were not cited in manuscript and were outsourced by web search, relied on interpretation from authors (may have differed slightly between different institutions within multi-site studies or clinicians' interpretation/execution of the protocol)

[a]Average return-to-play time of two groups within sport

[b]Total concussion time-loss was reported by 80 players, this figure is an average of these published data

[c]Data were changed by lead author from weeks [3.6 weeks missed (median, 95% CI 3.2–4.0.2.0)] to days

[d]Concussion return-to-play policy changed throughout this study, so this return-to-play policy duration is the average of all six years of the study, whilst the mandated period only appeared in the final year of the study (2009–2015)

female field-hockey, respectively [8]. Median RTP results support previous work [15, 50] suggesting SRC recovery may differ between specific sports, with clear trends of a consistently longer RTP time in rugby than in American football (18.5–34 days and 7–16 days, respectively). The frequent use of mandated sit-out periods in rugby internationally of > 10 days (as per Table 3) may be one confounding factor for this prolonged SRC recovery data. It should be noted, however, that longer RTP times in rugby within U.S. studies (3/8, 37.5%) were also reported. As rugby is not an official NCAA sport, other variables for concussion RTP time within these studies should be considered. Previous research suggests possible confounding factors for RTP time include varied access to medical care [51], SRC setting (training/match), team culture, and concussion education, all of which should be considered when evaluating published SRC RTP data [51, 52]. Other sports have more intra-sport than between-sport RTP duration variability, highlighting the potential impact of elements such as year of study, study design, and sample size on SRC recovery data. Interestingly, sports with mandated sit-out periods of 1 day or no same-day return still frequently resulted in over 2 times the RTP protocol duration, as seen in Table 3.

For several reasons, it is difficult to discern the full impact of extended mandated sit-out time on RTP duration. Firstly, only 7/58 (12.1%) datasets had more than a one-day mandated sit-out period, coupled with the highly individual nature of SRC and variety of SRC duration evident across all 58 datasets. Secondly, it is also challenging to truly grasp the effects of prolonged mandated sit-out periods or RTP protocols on SRC recovery times where compliance metrics are absent or differences in medical practitioner involvement are evident. While some researchers highlight compliance issues and/or mitigating mistakes (such as full completion of RTP protocol as an inclusion criterion), other compliance issues may be observed when comparing RTP protocols and/or mandated sit-out periods with RTP data. Two findings that highlight this are ladies gaelic football's median RTP duration reported at less than half of their RTP protocol [37], and the Wang et al. (2022) male soccer population reporting a median of 6.5 days RTP, whilst the corresponding RTP protocol was 7-days long [8]. It should be noted that this male soccer population had a very small sample size ($n = 4$), which may have impacted the findings [8].

Access to medical involvement or support staff within the studies varied, possibly affecting enforcement of and compliance with mandated sit-out periods and full completion of RTP protocols. Altogether, 15/19 (79.0%) of all eligible studies referenced medical practitioner involvement at some level (such as team doctor or athletic trainers). However, there was an observed trend of increased reference to medical involvement at some level of study within U.S. collegiate athletic populations (92.0%, $n = 11/12$) versus international populations (57.1%, $n = 4/7$). Lack of medical practitioner involvement and delayed access to specialty care have been previously associated with prolonged SRC recovery [53, 54], so these factors should be considered when evaluating SRC recovery research, as healthcare practitioner access and resources may vary substantially across contact field sports.

It proved difficult to decipher between policy guidelines versus mandated sit-out periods, where numerous policies required author interpretation. Likely, the individuality of SRC management [18] creates a barrier to universally mandated guidance. However, the ambiguity of language used, such as ''an athlete should'', may highlight an issue within the SRC RTP protocol framework where clarity should be offered on mandatory time away from sport versus suggestive frameworks; completion of the RTP protocol may be interpreted as optional rather than an obligation. It should be noted that some protocols included may now be outdated, such as NCAA guidelines now including mandatory requirements for medical personnel at contact sport events and mandatory completion of RTP protocols [55, 56], and the introduction of standardised SRC guidelines for all UK Grassroot Sports [57]. It is pertinent to the player's welfare that RTP protocols be followed and adequate recovery time attained to combat the risk of subsequent SRCs or musculoskeletal injury [58, 59], persisting SRC symptoms, and/or prolonged SRC recovery [3–5, 20].

The prevalence and/or incidence of SRC within the included contact field sports is difficult to determine due to the variation in SRC study design and lack of IR data. This may be because these papers focused on SRC recovery rather than SRC burden [60]. A lack of consistency is evident in the presentation of the available data, with the sample size often depicting several different sports, the inclusion of SRC-only data or all musculoskeletal injuries, and IR presentation varying in the inclusion of training/practice. When evaluating SRC research for use in clinical practice, it is important to account for these variables. Research definition inconsistencies should also be acknowledged when evaluating SRC publications, such as time-loss injuries being > 1 day or > 8 days away from sport [39, 43].

Limitations

This systematic review had several limitations. First, it was limited by the quality of eligible studies, with 6 studies indicating Level of Evidence 3 and 8 studies having a fair quality assessment. Next, the number of studies and/or data for each contact field sport varied, with only one eligible study including ladies gaelic football and sprint football, and others including small sample sizes of less than 10 in at least one sporting

code [8, 16, 30, 32, 40, 44], which typically occurred in multi-sport studies where concussion recovery time between sports was not the primary aim of the study. The use of SRC data from multiple sporting codes in the literature resulted in a lot of authors being contacted for sport-specific SRC data; frequently, the author did not respond to the request and thus their study was excluded. This, combined with some studies having variations in medical access for SRC diagnosis, management and treatment, alongside other resource availability, may be incomparable to other amateur populations, limiting the generalisability of our findings. Further, some of the included studies were limited by the lack of awareness of RTP protocol compliance, which may have affected the resulting RTP durations. Also, as RTP protocols had to be interpreted by the first author, there is a possibility of bias in their explanations in this review. Bias may also be found in included studies where the definition of 'amateur' sport was not established and perhaps inconsistent across studies.

Conclusion

SRC recovery time varied largely across amateur adult contact field sports, suggesting that sport type is an important consideration for healthcare providers involved in SRC management. The sport's required RTP protocol and mandated sit-out period may affect SRC recovery but should be considered alongside other confounding variables, such as access to medical practitioners and compliance with the RTP protocol when interpreting published SRC recovery metrics. Healthcare practitioners should be aware of the possible compliance issues in RTP protocols during SRC management to ensure full SRC recovery and safe RTP for all athletes. Future research should strive for common data elements within SRC recovery research to improve the comparability of studies, alongside recognition of medical involvement and/or treatment provision during the RTP process. Future enhanced diversity in study populations would allow for research to be generalisable globally.

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Declarations

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References

1. Harmon KG, Drezner JA, Gammons M, Guskiewicz KM, Halstead M, Herring SA et al (2013) American medical society for sports medicine position statement: concussion in sport. Br J Sports Med 47(1):15–26
2. Giza CC, Hovda DA (2001) The neurometabolic cascade of concussion. J Athl Train 36(3):228–235
3. Patricios JS, Schneider KJ, Dvorak J, Ahmed OH, Blauwet C, Cantu RC et al (2023) Consensus statement on concussion in sport: the 6th International Conference on Concussion in Sport–Amsterdam, October 2022. Br J Sports Med 57(11):695–711
4. McCrory P, Meeuwisse W, Aubry M, Cantu B, Dvorak J, Echemendia RJ et al (2013) Consensus statement on concussion in sport–the 4th International Conference on Concussion in Sport held in Zurich, November 2012. Clin J Sport Med 23(2):89–117
5. Broglio SP, McAllister T, Katz BP, LaPradd M, Zhou W, McCrea MA et al (2022) The natural history of Sport-Related concussion in collegiate athletes: findings from the NCAA-DoD CARE consortium. Sports Med Auckl NZ 52(2):403–415
6. Herring S, Kibler WB, Putukian M, Solomon GS, Boyajian-O'Neill L, Dec KL et al (2021) Selected issues in sport-related concussion (SRC|mild traumatic brain injury) for the team physician: a consensus statement. Br J Sports Med 55(22):1251–1261
7. Iverson GL, Gardner AJ, Terry DP, Ponsford JL, Sills AK, Broshek DK et al (2017) Predictors of clinical recovery from concussion: a systematic review. Br J Sports Med 51(12):941–948
8. Wang EX, Hwang CE, Nguyen JN, Segovia NA, Abrams GD, Kussman A (2022) Factors associated with a prolonged time to return to play after a concussion. Am J Sports Med 50(6):1695–1701
9. Wait TJ, Eck AG, Loose T, Drumm A, Kolaczko JG, Stevanovic O et al (2023) Median time to return to sports after concussion is within 21 days in 80% of published studies. Arthroscopy: The Journal of Arthroscopic & Related Surgery 39(3):887–901
10. Shirley HL, Walker B, Reneker JC (2024) University club sport student-athlete concussion profiles differ by sport and sex. Recreat Sports J 48(1):66–74
11. Walshe A, Daly E, Ryan L (2022) Epidemiology of sport-related concussion rates in female contact/collision sport: a systematic review. BMJ Open Sport — Exerc Med 8(3):e001346
12. Pierpoint LA, Collins C (2021) Epidemiology of sport-related concussion. Clin Sports Med 40(1):1–18
13. Van Pelt KL, Puetz T, Swallow J, Lapointe AP, Broglio SP (2021) Data-driven risk classification of concussion rates: a systematic review and meta-analysis. Sports Med 51(6):1227–1244
14. Matias-Soto J, Infante-Cano M, García-Muñoz C, Pineda-Escobar S, Martinez-Calderon J (2024) Concussion incidence by type of sport: differences by sex, age groups, type of session, and level of play an overview of systematic reviews with meta-analysis. J Orthop Sports Phys Ther 54(11):1–9
15. Liebel SW, Van Pelt KL, Pasquina PF, McAllister TW, McCrea MA, Broglio SP et al (2023) Sport-specific recovery trajectories for NCAA collegiate athletes following concussion. Ann Biomed Eng. https://doi.org/10.1007/s10439-023-03406-8
16. Davis-Hayes C, Gossett JD, Levine WN, Shams T, Harada J, Mitnick J et al (2017) Sex-specific outcomes and predictors of concussion recovery. J Am Acad Orthop Surg 25(12):818–828

17. Broglio SP, Cantu RC, Gioia GA, Guskiewicz KM, Kutcher J, Palm M et al (2014) National Athletic Trainers' Association position statement: management of sport concussion. J Athl Train 49(2):245–65
18. Scorza KA, Cole W (2019) Current concepts in concussion: initial evaluation and management. Am Fam Physician 99(7):426–434
19. Beakey M, Blake C, Tiernan S, Collins K (2016) Are Irish sporting organisations neglecting the safety of their athletes? A narrative review on concussion Return-to-Play protocols. Eur J Sports Med 3
20. Prock M, O'Sullivan DM, Tiernan S (2024) Comparing return to play protocols after sports-related concussion among international sporting organisations. Phys Sportsmed 0(0):1–11
21. Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD et al (2021) The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ. n71
22. Ardern CL, Büttner F, Andrade R, Weir A, Ashe MC, Holden S et al (2022) Implementing the 27 PRISMA 2020 statement items for systematic reviews in the sport and exercise medicine, musculoskeletal rehabilitation and sports science fields: the persist (implementing Prisma in Exercise, Rehabilitation, sport medicine and sports science) guidance. Br J Sports Med 56(4):175–195
23. Rethlefsen ML, Farrell AM, Osterhaus Trzasko LC, Brigham TJ (2015) Librarian co-authors correlated with higher quality reported search strategies in general internal medicine systematic reviews. J Clin Epidemiol 68(6):617–626
24. EndNote [Internet] Clarivate. [cited 2024 Jun 7]. Available from: https://clarivate.com/innovation-exchange/solution/endnote/
25. Covidence - Better systematic review management [Internet]. Covidence. [cited 2024 Jun 7]. Available from: https://www.covidence.org/
26. Rice SG (2008) Medical conditions affecting sports participation. Pediatrics 121(4):841–848
27. Van Pelt KL, Allred D, Cameron KL, Campbell DE, D'Lauro CJ, He X et al (2019) A cohort study to identify and evaluate concussion risk factors across multiple injury settings: findings from the CARE consortium. Inj Epidemiol 6(1):1
28. Study Quality Assessment Tools | NHLBI NIH [Internet]. [cited 2024 Jun 27]. Available from: https://www.nhlbi.nih.gov/health-topics/study-quality-assessment-tools
29. Ebell MH, Siwek J, Weiss BD, Woolf SH, Susman J, Ewigman B et al (2004) Strength of recommendation taxonomy (SORT): A Patient-Centered approach to grading evidence in the medical literature. Am Fam Physician 69(3):548–556
30. Howell DR, Brilliant AN, Oldham JR, Berkstresser B, Wang F, Meehan WP (2020) Exercise in the first week following concussion among collegiate athletes: preliminary findings. J Sci Med Sport 23(2):112–117
31. Sterling J, Robinson S, Meyers M, Paper (2023) Effect of mouthguard type on severity of concussion during Game-Related college football Competition. A 16-Season prospective study. Orthop J Sports Med 60(7 suppl3):2325967123S00085
32. Asken BM, McCrea MA, Clugston JR, Snyder AR, Houck ZM, Bauer RM (2016) Playing through it: delayed reporting and removal from athletic activity after concussion predicts prolonged recovery. J Athl Train 51(4):329–335
33. Bayram JM, Hamilton DF, Saunders DH (2020) Epidemiology of American football injuries at universities in the United Kingdom. Orthop J Sports Med 8(10):2325967120960206
34. Wasserman EB, Kerr ZY, Zuckerman SL, Covassin T (2016) Epidemiology of sports-related concussions in National collegiate athletic association athletes from 2009–2010 to 2013–2014: symptom prevalence, symptom resolution time, and return-to-play time. Am J Sports Med 44(1):226–233
35. Bretzin AC, Esopenko C, D'Alonzo BA, Wiebe DJ (2022) Clinical recovery timelines after sport-related concussion in men's and women's collegiate sports. J Athl Train 57(7):678–687
36. Covassin T, Moran R, Elbin RJ (2016) Sex differences in reported concussion injury rates and time loss from participation: an update of the National collegiate athletic association injury surveillance program from 2004–2005 through 2008–2009. J Athl Train 51(3):189–194
37. Leahy R, Rochfort KD, Whyte E, Kontos AP, Collins MW, O'Connor S (2023) Concussion in ladies Gaelic football: self-reported history, clinical profiles, and management behavior. Clin J Sport Med 33(2):157
38. Yener N, Glendon K, Pain MTG (2022) The association between sport-related concussion and musculoskeletal injury in university rugby athletes. Phys Ther Sport 55:264–270
39. Yeomans C, Kenny IC, Cahalan R, Warrington GD, Harrison AJ, Purtill H et al (2021) Injury trends in Irish amateur rugby: an epidemiological comparison of men and women. Sports Health 13(6):540–547
40. Gallagher V, Kramer N, Abbott K, Alexander J, Breiter H, Herrold A et al (2018) The effects of sex differences and hormonal contraception on outcomes after collegiate Sports-Related concussion. J Neurotrauma 35(11):1242–1247
41. Ball S, Halaki M, Orr R (2020) Movement demands and injury characteristics in Under-20-Years university rugby union players. J Athl Train 55(4):376–383
42. Fuller CW, Taylor A, Raftery M (2018) Eight-season epidemiological study of injuries in men's international Under-20 rugby tournaments. J Sports Sci 36(15):1776–1783
43. Roberts SP, Trewartha G, England M, Goodison W, Stokes KA (2016) Concussions and head injuries in English community rugby union match play. Am J Sports Med 45(2):480–487
44. Buckley TA, Munkasy BA, Evans KM, Clouse B (2022) Acute physical and mental activity influence on concussion recovery. Med Sci Sports Exerc 54(2):307–312
45. Beidler E, Kelshaw PM, Wallace J, Larson MJ, Munce TA, Donahue CC et al (2023) Racial identity and concussion diagnosis and recovery trajectories in collegiate athletes: a LIMBIC MATARS investigation. Brain Inj, pp 1–9
46. McCrory P, Meeuwisse W, Dvorak J, Aubry M, Bailes J, Broglio S, et al (2017) Consensus statement on concussion in sport—the 5th international conference on concussion in sport held in Berlin, October 2016. Br J Sports Med 51(11):838–47
47. McCrory P, Meeuwisse W, Johnston K, Dvorak J, Aubry M, Molloy M et al (2009) Consensus statement on concussion in sport: the 3rd international conference on concussion in sport held in Zurich, November 2008. J Athl Train 44(4):434–48
48. Buckley TA, Baugh CM, Meehan WP, DiFabio MS (2017) Concussion Management Plan Compliance: A Study of NCAA Power 5 Conference Schools. Orthop J Sports Med 5(4):2325967117702606
49. Irish Rugby | IRFU Concussion Protocols [Internet] [cited 2024 Jun 10]. Available from: https://www.irishrugby.ie/playing-the-game/medical/irfu-concussion-protocols/
50. Nelson LD, Tarima S, LaRoche AA, Hammeke TA, Barr WB, Guskiewicz K et al (2016) Preinjury somatization symptoms contribute to clinical recovery after sport-related concussion. Neurology 86(20):1856–1863
51. Putukian M, Aubry M, McCrory P (2009) Return to play after sports concussion in elite and non-elite athletes? Br J Sports Med 43(Suppl 1):i28–31
52. Corman SR, Adame BJ, Tsai JY, Ruston SW, Beaumont JS, Kamrath JK et al (2019) Socioecological influences on concussion reporting by NCAA division 1 athletes in high-risk sports. PLoS One 14(5):e0215424

53. Putukian M, Purcell L, Schneider KJ, Black AM, Burma JS, Chandran A et al (2023) Clinical recovery from concussion–return to school and sport: a systematic review and meta-analysis. Br J Sports Med 57(12):798–809
54. Kontos AP, Jorgensen-Wagers K, Trbovich AM, Ernst N, Emami K, Gillie B et al (2020) Association of time since injury to the first clinic visit with recovery following concussion. JAMA Neurol 77(4):435–440
55. NCAA. Concussion Safety Protocol Checklist [Internet] (2023) Available from: https://ncaaorg.s3.amazonaws.com/ssi/concussion/SSI_ConcussionSafetyProtocolChecklist.pdf
56. Updated NCAA, Concussion Protocol Requirements and Compliance with the Arrington Settlement Agreement - Law Firm Alliance [Internet]. [cited 2024 Jul 11]. Available from: https://www.lawfirmalliance.org/news-insights-events/updated-ncaa-concussion-protocol-requirements-and-compliance-with-the-arrington-settlement-agreement
57. UK Government. UK Concussion Guidelines for Non-Elite (Grassroots) Sport [Internet] (2023) Available from: https://sramedia.s3.amazonaws.com/media/documents/9ced1e1a-5d3b-4871-9209-bff4b2575b46.pdf
58. McPherson AL, Nagai T, Webster KE, Hewett TE (2019) Musculoskeletal injury risk after sport-related concussion: a systematic review and meta-analysis. Am J Sports Med 47(7):1754–1762
59. McCrea M, Broglio S, McAllister T, Zhou W, Zhao S, Katz B et al (2020) Return to play and risk of repeat concussion in collegiate football players: comparative analysis from the NCAA concussion study (1999–2001) and CARE consortium (2014–2017). Br J Sports Med 54(2):102–109
60. Verhagen E, Clarsen B, Van Der Graaff L, Bahr R (2024) Do not neglect injury severity and burden when assessing the effect of sports injury prevention interventions: time to paint the whole picture. Br J Sports Med bjsports-2024-108215
61. GAA. Concussion Management Guideliness for Gaelic Games 2018 [Internet] (2018) Available from: https://learning.gaa.ie/sites/default/files/Concussion%20Management%20Guidelines%20for%20Gaelic%20Games%20November%202018_0.pdf
62. Report a Concern Concussion Management | Rugby Australia [Internet]. [cited 2024 Jun 10]. Available from: https://australia.rugby/about/codes-and-policies/safety-and-welfare/concussion-management
63. worldrugby.org Concussion Guidance | World Rugby [Internet]. [cited 2024 Jun 10]. Available from: https://www.world.rugby/the-game/player-welfare/medical/concussion/concussion-guidelines
64. Glendon K, Blenkinsop G, Belli A, Pain M (2021) Prospective study with specific re-assessment time points to determine time to recovery following a sports-related concussion in university-aged student-athletes. Phys Ther Sport 52:287–296

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